undertaking of space related activities, Satellite Telecommunications, and domestic regulation and development of Remote Sensing Activities.

The book concludes with a Chapter on international and domestic regulation of Intellectual Property rights of Inventions made in outer space. Given the interesting topics and a well organized and written text, persons with Spanish fluency who could benefit from the more in-depth Spanish text would likely find this book a useful tool in understanding the current framework and issues in Space Law.

Michael A. Gorove
Attorney at Law
Associate Editor, J. SPACE L.

SPACE COOPERATION INTO THE 21ST CENTURY, edited by Peter M. Bainum, Gayle L. May, Makoto Nagatomo, Kuninori T. Uesugi, Fu Bingchen and Zhang Hui (Am. Astronautical Soc'y, Advances in the Astronautical Sciences, vol. 96, Univelt 1997), pp. xviii, 1080.

SPACE SAFETY AND RESCUE 1995, edited by Gloria W. Heath (Am. Astronautical Soc'y, Science and Technology Ser., vol. 93, Univelt 1997), pp. xii, 469.

The first of these two publications of the American Astronautical Society includes papers presented at the Seventh International Space Conference of Pacific-Basin Societies held July 15-18, 1997, in Nagasaki, Japan. Its theme focused on "Space Cooperation into the 21st Century". Among the variety of topics addressed at the technical sessions were space communications, space transportation and propulsion, space science and missions, and beneficial applications of space systems. The technical papers were preceded by sections addressing national and international programs and contributions submitted by international students for a competition in connection with the conference. Sections which may be of special interest to lawyers include those dealing with space debris and environmental issues, as well as those focusing on manned space flight, the international space station and a Pacific Spaceport.

While the second publication under the auspices of the International Academy of Astronautics, as those under "Space Safety and Rescue" title in prior years, follows its traditional pattern by focusing on the technical aspects of the subject matter, it also provides data on effects and ramifications involving risk assessment and management issues in different scenarios. Random examples of potential legal interest may be found in both the engineering ("Space Debris Mitigation" by Joseph P. Loftus, Jr.) and the economical aspects of space debris mitigation measures ("Meteoroid and Orbital Debris Risk Mitigation for the International Space Station" by Allen J. Lindenmoyer, *et al.*). Unfortunately, the publication has no index and a perfunctory glance reveals no general exposition of safety concerns about earth threatening asteroids, perhaps because this had not been a major news item in 1995.

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* Excerpts of the Agreement, are reproduced in CURRENT DOCUMENTS, *infra*. For full text, see UNITED STATES SPACE LAW - NATIONAL AND INTERNATIONAL REGULATION, Sec. II. 22(f) (Stephen Gorove ed., OCEANA 1998).

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CURRENT DOCUMENTS

AGREEMENT AMONG THE GOVERNMENT OF CANADA, GOVERNMENTS OF MEMBER STATES OF THE EUROPEAN SPACE AGENCY, THE GOVERNMENT OF JAPAN, THE GOVERNMENT OF THE RUSSIAN FEDERATION, AND THE GOVERNMENT OF THE UNITED STATES OF AMERICA CONCERNING COOPERATION ON THE CIVIL INTERNATIONAL SPACE STATION.

Done at Washington, January 29, 1998

Excerpts*

Article 1
Object and Scope

1. The object of this Agreement is to establish a long-term international cooperative framework among the Partners, on the basis of genuine partnership, for the detailed design, development, operation, and utilization of a permanently inhabited civil international Space Station for peaceful purposes, in accordance with international law. This civil international Space Station will enhance the scientific, technological, and commercial use of outer space. This Agreement specifically defines the civil international Space Station program and the nature of this partnership, including the respective rights and obligations of the Partners in this cooperation. This Agreement further provides for mechanisms and arrangements designed to ensure that its object is fulfilled.

2. The Partners will join their efforts, under the lead role of the United States for overall management and coordination, to create an integrated international Space Station. The United States and Russia, drawing on their extensive experience in human space flight, will produce elements which serve as the foundation for the international Space Station. The European Partner and Japan will produce elements that will significantly enhance the Space Station's capabilities. Canada's contribution will be an essential part of the Space Station. This Agreement lists in the Annex the elements to be provided by the Partners to form the international Space Station.

3. The permanently inhabited civil international Space Station (hereinafter "the Space Station") will be a multi-use facility in low-earth orbit, with flight elements and Space Station-unique ground elements provided by all the Partners. By providing Space Station flight elements, each Partner acquires certain rights to use the Space Station and participates in its management in accordance with this Agreement, the MOUs, and implementing arrangements.

4. The Space Station is conceived as having an evolutionary character. The Partner States' rights and obligations regarding evolution shall be subject to specific provisions in accordance with Article 14.

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* For full texts of the Agreement and the Memorandum of Understanding of the same date, see UNITED STATES SPACE LAW - NATIONAL AND INTERNATIONAL REGULATION, Secs. II. 22 (f) and (g) (Stephen Gorove ed., OCEANA 1998).

Article 2
International Rights and Obligations

1. The Space Station shall be developed, operated, and utilized in accordance with international law, including the Outer Space Treaty, the Rescue Agreement, the Liability Convention, and the Registration Convention.

2. Nothing in this Agreement shall be interpreted as:

(a) modifying the rights and obligations of the Partner States found in the treaties listed in paragraph 1 above, either toward each other or toward other States, except as otherwise provided in Article 16;

(b) affecting the rights and obligations of the Partner States when exploring or using outer space, whether individually or in cooperation with other States, in activities unrelated to the Space Station; or

(c) constituting a basis for asserting a claim to national appropriation over outer space or over any portion of outer space.

Article 3
Definitions

For the purposes of this Agreement, the following definitions shall apply:

(a) "this Agreement":
the present Agreement, including the Annex;

(b) "the Partners" (or, where appropriate, "each Partner"):
the Government of Canada; the European Governments listed in the Preamble which become parties to this Agreement, as well as any other European Government that may accede to this Agreement in accordance with Article 25(3), acting collectively as one Partner; the Government of Japan; the Government of the Russian Federation; and the Government of the United States;

(c) "Partner State":
each Contracting Party for which this Agreement has entered into force, in accordance with Article 25.

Article 4
Cooperating Agencies

1. The Partners agree that the Canadian Space Agency (hereinafter "CSA") for the Government of Canada, the European Space Agency (hereinafter "ESA") for the European Governments, the Russian Space Agency (hereinafter "RSA") for Russia, and the National Aeronautics and Space Administration (hereinafter "NASA") for the United States shall be the Cooperating Agencies responsible for implementing Space Station cooperation. The

Government of Japan's Cooperating Agency designation for implementing Space Station cooperation shall be made in the Memorandum of Understanding between NASA and the Government of Japan referred to in paragraph 2 below.

2. The Cooperating Agencies shall implement Space Station cooperation in accordance with the relevant provisions of this Agreement, the respective Memoranda of Understanding (MOUs) between NASA and CSA, NASA and ESA, NASA and the Government of Japan, and NASA and RSA concerning cooperation on the civil international Space Station, and arrangements between or among NASA and the other Cooperating Agencies implementing the MOUs (implementing arrangements). The MOUs shall be subject to this Agreement, and the implementing arrangements shall be consistent with and subject to the MOUs.

3. Where a provision of an MOU sets forth rights or obligations accepted by a Cooperating Agency (or, in the case of Japan, the Government of Japan) not a party to that MOU, such provision may not be amended without the written consent of that Cooperating Agency (or, in the case of Japan, the Government of Japan).

Article 5
Registration; Jurisdiction and Control

1. In accordance with Article II of the Registration Convention, each Partner shall register as space objects the flight elements listed in the Annex which it provides, the European Partner having delegated this responsibility to ESA, acting in its name and on its behalf.

2. Pursuant to Article VIII of the Outer Space Treaty and Article II of the Registration Convention, each Partner shall retain jurisdiction and control over the elements it registers in accordance with paragraph 1 above and over personnel in or on the Space Station who are its nationals. The exercise of such jurisdiction and control shall be subject to any relevant provisions of this Agreement, the MOUs, and implementing arrangements, including relevant procedural mechanisms established therein.

Article 6
Ownership of Elements and Equipment

1. Canada, the European Partner, Russia, and the United States, through their respective Cooperating Agencies, and an entity designated by Japan at the time of the deposit of its instrument under Article 25(2), shall own the elements listed in the Annex that they respectively provide, except as otherwise provided for in this Agreement. The Partners, acting through their Cooperating Agencies, shall notify each other regarding the ownership of any equipment in or on the Space Station.

2. The European Partner shall entrust ESA, acting in its name and on its behalf, with ownership over the elements it provides, as well as over any other equipment developed and funded under an ESA programme as a contribution to the Space Station, its operation or utilization.

3. The transfer of ownership of the elements listed in the Annex or of equipment in or on the Space Station shall not affect the rights and obligations of the Partners under this Agreement, the MOUs, or implementing arrangements.

4 Equipment in or on the Space Station shall not be owned by, and ownership of elements listed in the Annex shall not be transferred to, any non-Partner or private entity under the jurisdiction of a non-Partner without the prior concurrence of the other Partners. Any transfer of ownership of any element listed in the Annex shall require prior notification of the other Partners.

5. The ownership of equipment or material provided by a user shall not be affected by the mere presence of such equipment or material in or on the Space Station.

6. The ownership or registration of elements or the ownership of equipment shall in no way be deemed to be an indication of ownership of material or data resulting from the conduct of activities in or on the Space Station.

7. The exercise of ownership of elements and equipment shall be subject to any relevant provisions of this Agreement, the MOUs, and implementing arrangements, including relevant procedural mechanisms established therein.

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Article 16
Cross-Waiver of Liability

1. The objective of this Article is to establish a cross-waiver of liability by the Partner States and related entities in the interest of encouraging participation in the exploration, exploitation, and use of outer space through the Space Station. This cross-waiver of liability shall be broadly construed to achieve this objective.

2. For the purposes of this Article:

(a) A "Partner State" includes its Cooperating Agency. It also includes any entity specified in the MOU between NASA and the Government of Japan to assist the Government of Japan's Cooperating Agency in the implementation of that MOU.

(b) The term "related entity" means:

(1) a contractor or subcontractor of a Partner State at any tier;

(2) a user or customer of a Partner State at any tier; or

(3) a contractor or subcontractor of a user or customer of a Partner State at any tier.

This subparagraph may also apply to a State, or an agency or institution of a State, having the same relationship to a Partner State as described in subparagraphs 2(b)(1) through 2(b)(3) above or otherwise engaged in the implementation of Protected Space Operations as defined in subparagraph 2 (f) below.

"Contractors" and "subcontractors" include suppliers of any kind.

(c) The term "damage" means:

(1) bodily injury to, or other impairment of health of, or death of, any person;

(2) damage to, loss of, or loss of use of any property;

(3) loss of revenue or profits; or

(4) other direct, indirect or consequential damage.

(d) The term "launch vehicle" means an object (or any part thereof) intended for launch, launched from Earth, or returning to Earth which carries payloads or persons, or both.

(e) The term "payload" means all property to be flown or used on or in a launch vehicle or the Space Station.

(f) The term "Protected Space Operations" means all launch vehicle activities, Space Station activities, and payload activities on Earth, in outer space, or in transit between Earth and outer space in implementation of this Agreement, the MOUs, and implementing arrangements. It includes, but is not limited to:

(1) research, design, development, test, manufacture, assembly, integration, operation, or use of launch or transfer vehicles , the Space Station, or a payload, as well as related support equipment and facilities and services; and

(2) all activities related to ground support, test, training, simulation, or guidance and control equipment and related facilities or services.

"Protected Space Operations" also includes all activities related to evolution of the Space Station, as provided for in Article 14. "Protected Space Operations" excludes activities on Earth which are conducted on return from the Space Station to develop further a payload's product or process for use other than for Space Station related activities in implementation of this Agreement.

3. (a) Each Partner State agrees to a cross-waiver of liability pursuant to which each Partner State waives all claims against any of the entities or persons listed in subparagraphs 3(a)(1) through 3(a)(3) below based on damage arising out of Protected Space Operations. This cross-waiver shall apply only if the person, entity, or property causing the damage is involved in Protected Space Operations and the person, entity, or property damaged is damaged by virtue of its involvement in Protected Space Operations. The cross-waiver shall apply to any claims for damage, whatever the legal basis for such claims against:

(1) another Partner State;

(2) a related entity of another Partner State;

(3) the employees of any of the entities identified in subparagraphs 3(a)(1) and 3(a)(2) above.

(b) In addition, each Partner State shall, by contract or otherwise, extend the cross-waiver of liability as set forth in subparagraph 3(a) above to its related entities by requiring them to:

(1) waive all claims against the entities or persons identified in subparagraphs 3(a)(1) through 3(a)(3) above; and

(2) require that their related entities waive all claims against the entities or persons identified in subparagraphs 3(a)(1) through 3(a)(3) above.

(c) For avoidance of doubt, this cross-waiver of liability includes a cross-waiver of liability arising from the Liability Convention where the person, entity, or property causing the damage is involved in Protected Space Operations and the person, entity, or property damaged is damaged by virtue of its involvement in Protected Space Operations.

(d) Notwithstanding the other provisions of this Article, this cross-waiver of liability shall not be applicable to:

(1) claims between a Partner State and its related entity or between its own related entities;

(2) claims made by a natural person, his/her estate, survivors or subrogees (except when a subrogee is a Partner State) for bodily injury to, or other impairment of health of, or death of such natural person;

(3) claims for damage caused by willful misconduct;

(4) intellectual property claims;

(5) claims for damage resulting from a failure of a Partner State to extend the cross-waiver of liability to its related entities, pursuant to subparagraph 3(b) above.

(e) With respect to subparagraph 3(d)(2) above, in the event that a subrogated claim of the Government of Japan is not based upon government employee accident compensation law, the Government of Japan shall fulfill its obligation to waive such subrogated claim by ensuring that any assisting entity specified pursuant to subparagraph 2(a) above indemnifies, in a manner consistent with Article 15(2) and in accordance with applicable laws and regulations of Japan, any entity or person identified in subparagraphs 3(a)(1) through 3(a)(3) above against liability arising from such subrogated claim by the Government of Japan. Nothing in this Article shall preclude the Government of Japan from waiving the foregoing subrogated claims.

(f) Nothing in this Article shall be construed to create the basis for a claim or suit where none would otherwise exist.